



Did you Know? VERIFI Tips & Tricks



U.S. DEPARTMENT
of **ENERGY**

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FOR THE US DEPARTMENT OF ENERGY



Developers Console

- How to open

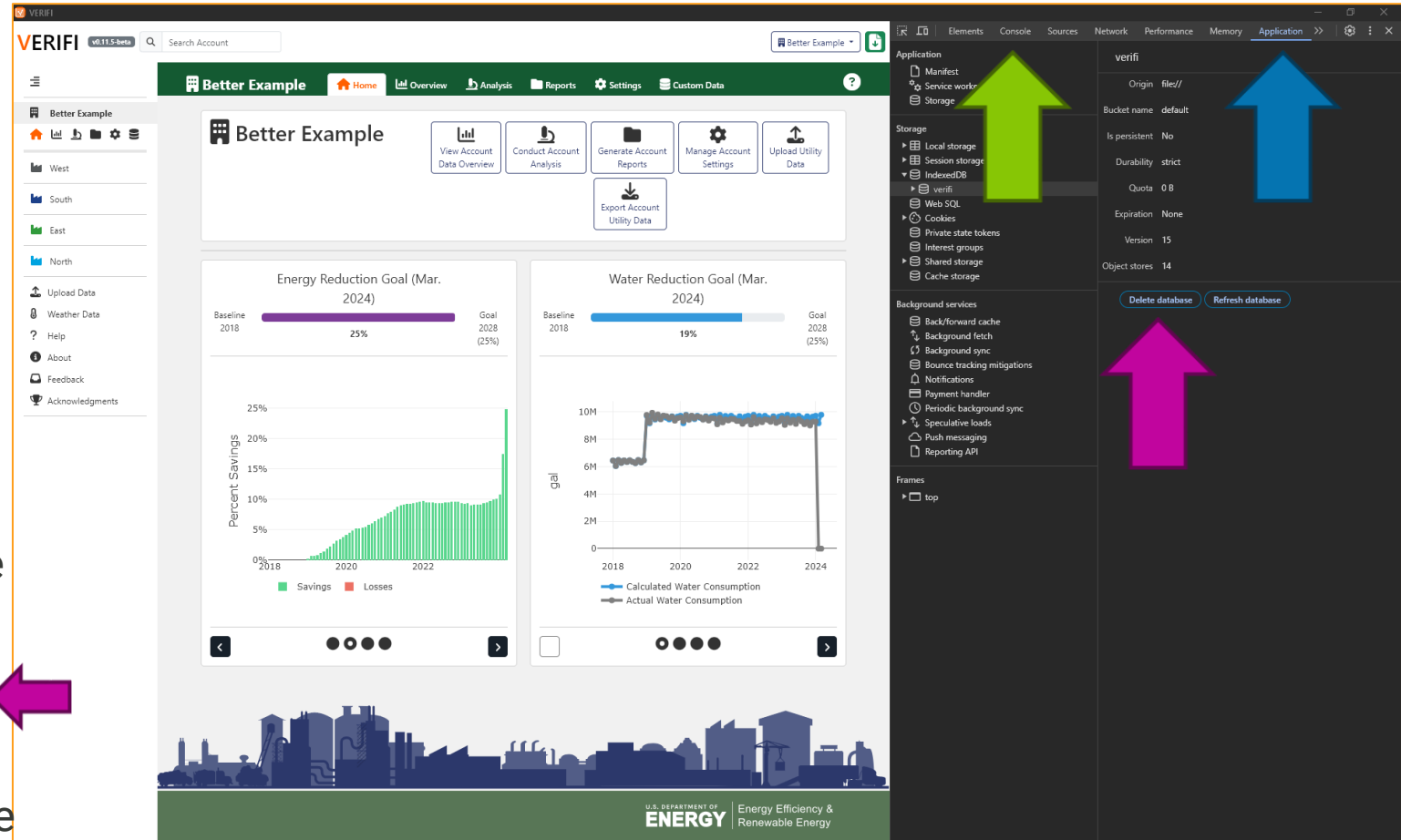
- Online: [ctrl]+[sft]+[i]
F12
- Download: [ctrl]+[i]

- What is it for?

➡ Console: Will show any errors that are occurring (red text)

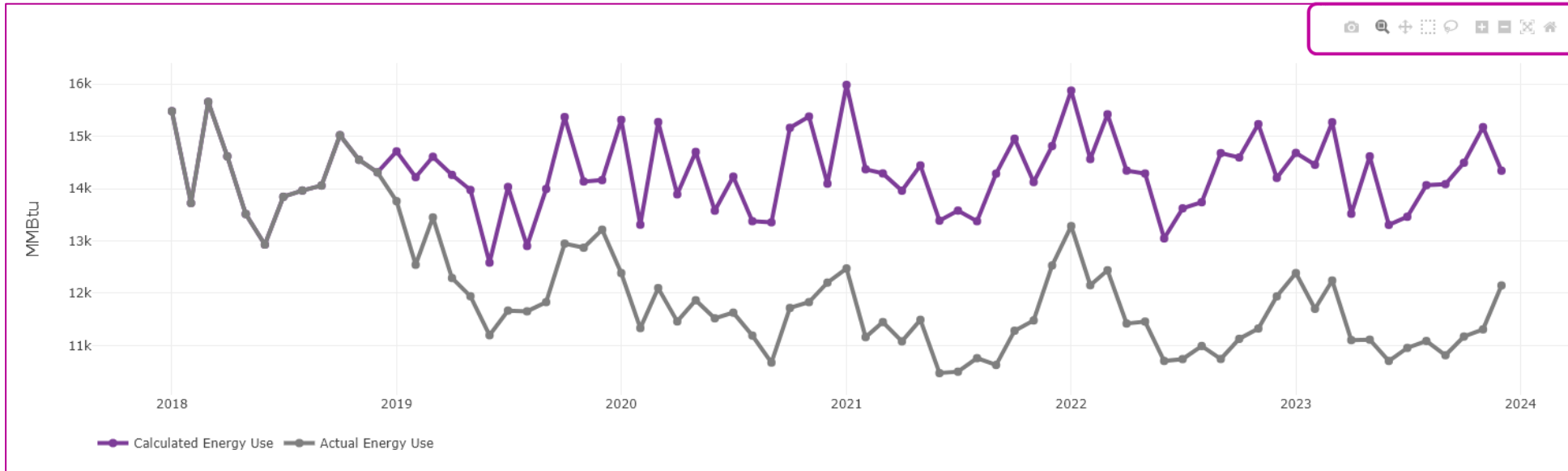
➡ Application: Can delete everything in application REALLY fast

- Hard Refresh: force browser to update application (right click the normal refresh button)



Copy & Control Graphs

Look for the Plotly menu in the corner of graphs

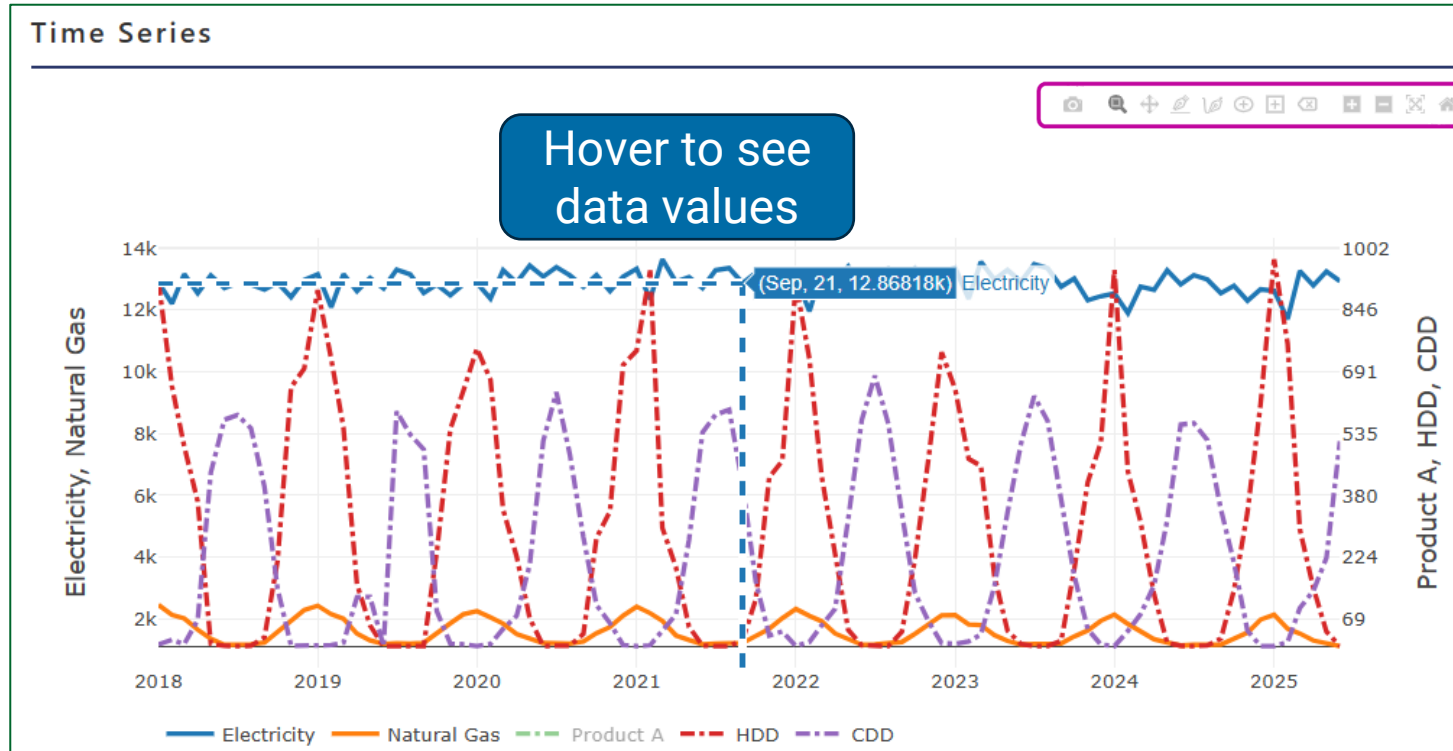


1 2 3 4 5 6 7

- 1 Save graph as png
- 2 Click & Drag to zoom
- 3 Click & Drag to move axis
- 4 Click & Drag to highlight data
- 5 Manual zoom in/out
- 6 Autoscale
- 7 Reset Axes (or double click)

Copy & Control Graphs

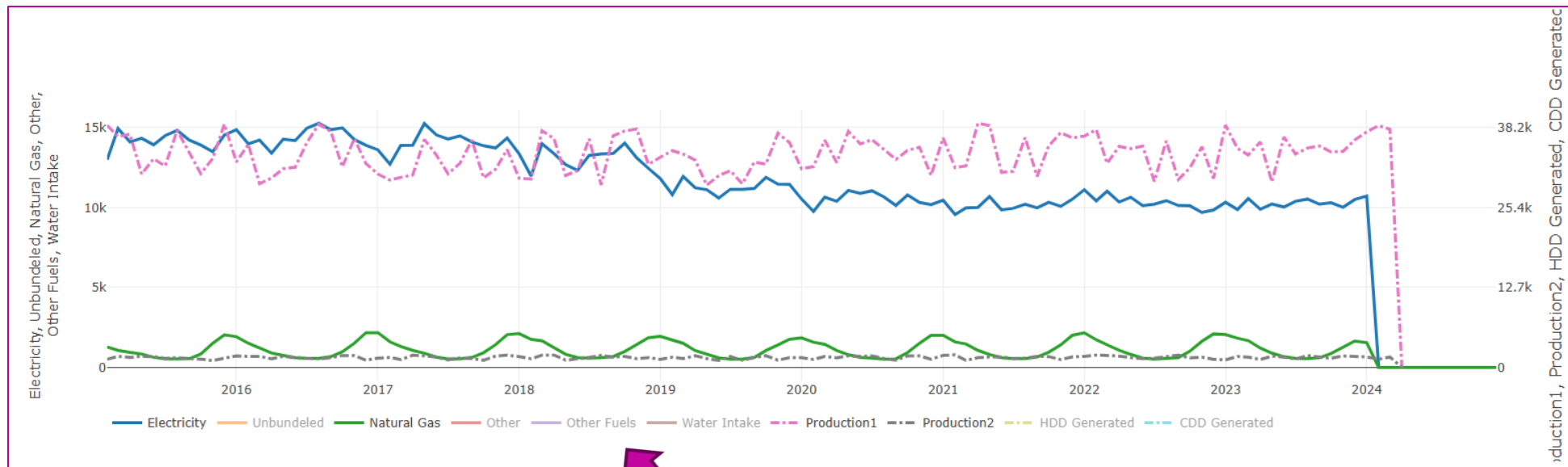
Look for the Plotly menu in the corner of graphs



- 1 Save graph as png
- 2 Click & Drag to zoom (default)
- 3 Click & Drag to move axis
- 4 Draw a line
- 5 Draw free freeform
- 6 Draw an oval
- 7 Draw a rectangle
- 8 Delete active shape
- 9 Manual zoom in/out
- 10 Autoscale
- 11 Reset Axes (or double click)

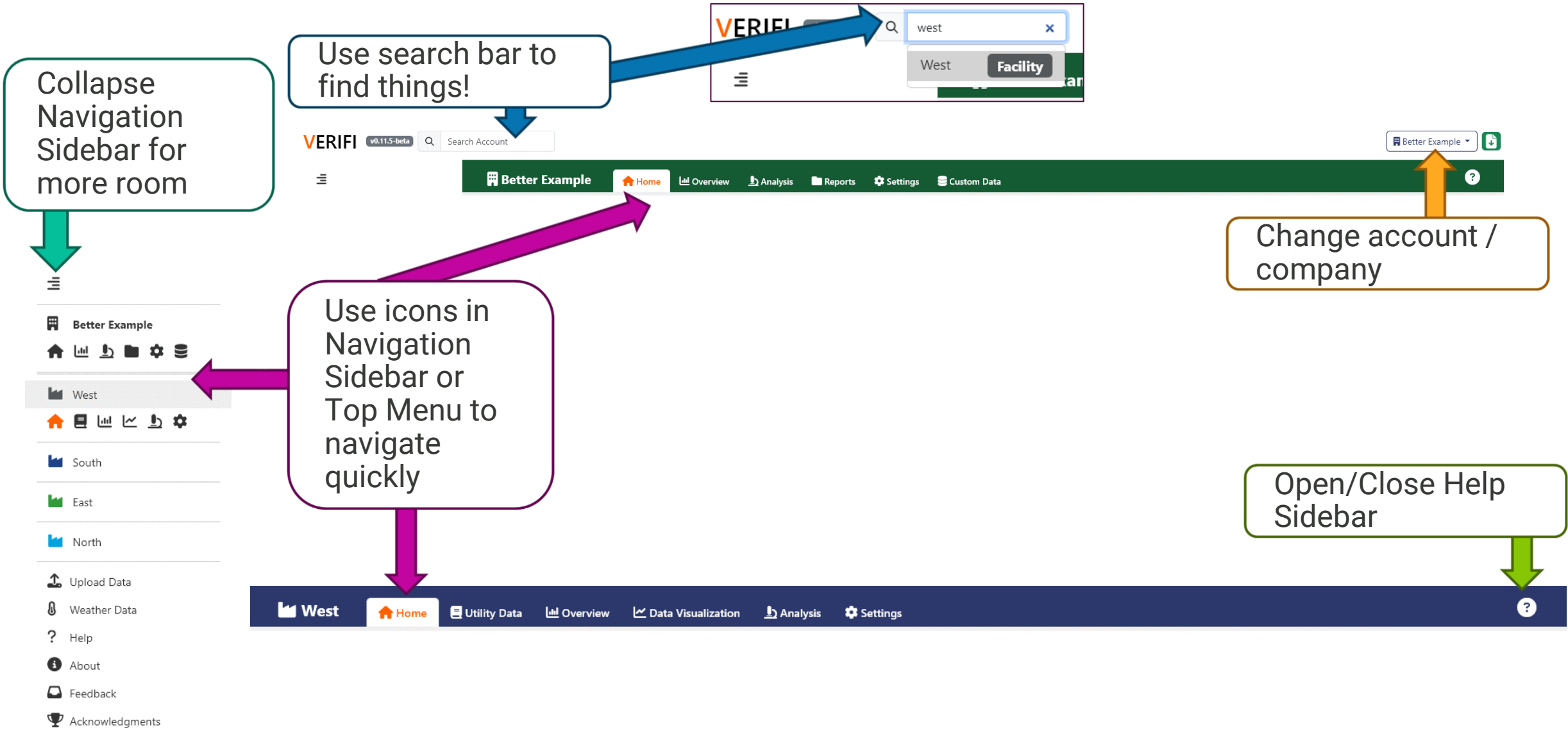
Control Data within Graphs

- Click legend items to hide
- Double click legend items to hide all except that item



Hide data easily

Faster Navigation



Better Example
Home
Overview
Analysis
Reports
Settings
Custom Data
?

Report Setup
Better Plants Report
All Account Reports

Annual Reporting Form

BETTER BUILDINGS, BETTER PLANTS

GENERATED BY VERIFI

Better Example

Contact Information

Name: Thomas Wenning
Phone: 8655559437
Email: tom.wenning@example.com

	Baseline Year 2018	Report Year 2023
Number of Participating Plants*	3	4
Primary Energy Consumed (MMBtu)		
Electricity	347,454	355,594
Natural Gas	45,197	69,043
Total Primary Energy Consumed, (MMBtu)	392,661	424,637
Weather/Production/Other Normalizing related Adjustment for Baseline Primary Energy, (+/- MMBtu)	881.62	
Baseline Adjustment Due to Increase/Decrease in the Number of Facilities Reporting Relative to Baseline Year or Other Operational Changes, (+/- MMBtu)	78,988	
Adjusted Baseline of Primary Energy (MMBtu)	472,531	
New Energy Savings for Current Year (MMBtu)		1,107.7
Total Energy Savings since Baseline Year (MMBtu)		47,894
Annual Improvement In Energy Intensity for Current Year (%)**		0.46 %
Total Improvement In Energy Intensity (%)**		10.14 %

*Participating plants should be only include those located in the United States

**Please refer to the DOE's Energy Baseline Guidance document to determine changes in intensity. Improvement in performance should be reported as a positive number.

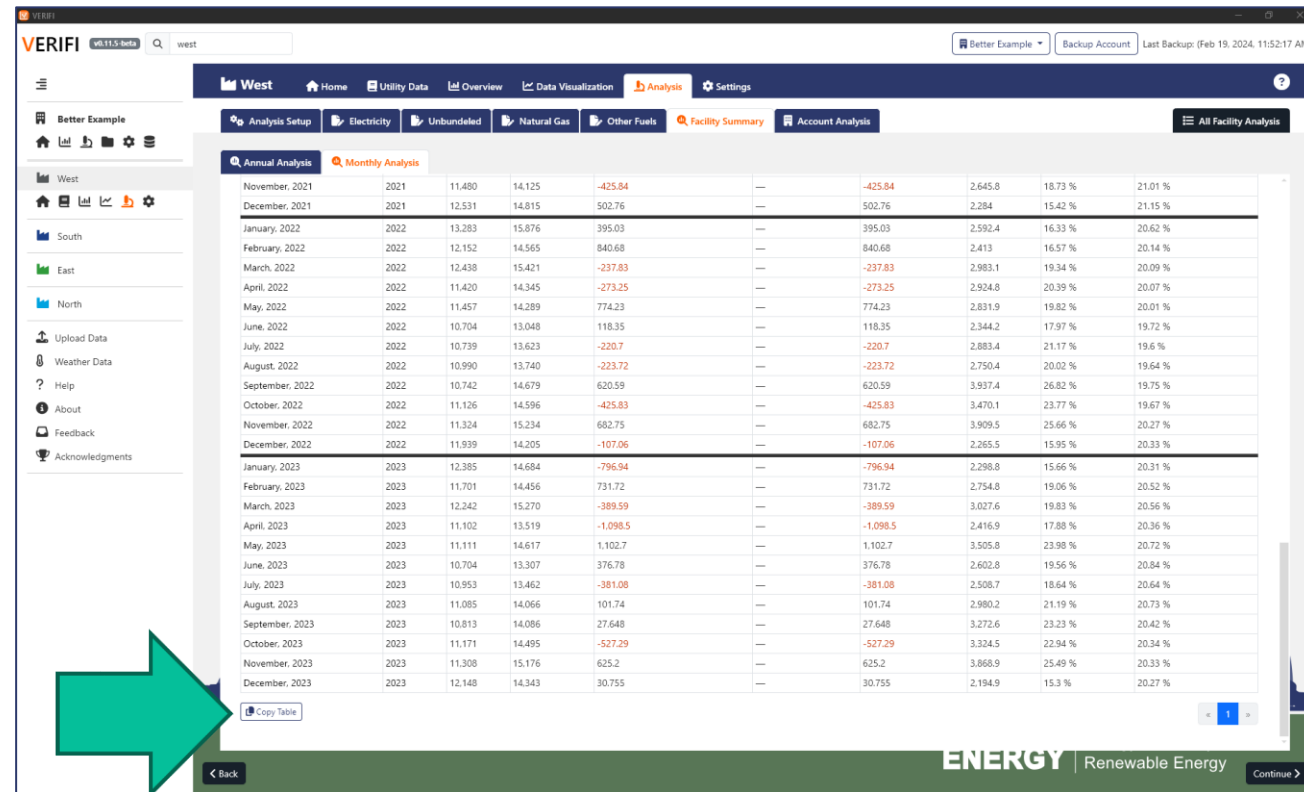
Please describe any methods undertaken to normalize energy intensity data or adjust baseline data to account for economic and other factors that affect energy use:

DRAFT



Copy Tables

- Want to play with data in Excel because something seems wrong? → Look for the “Copy Table” Button below the table

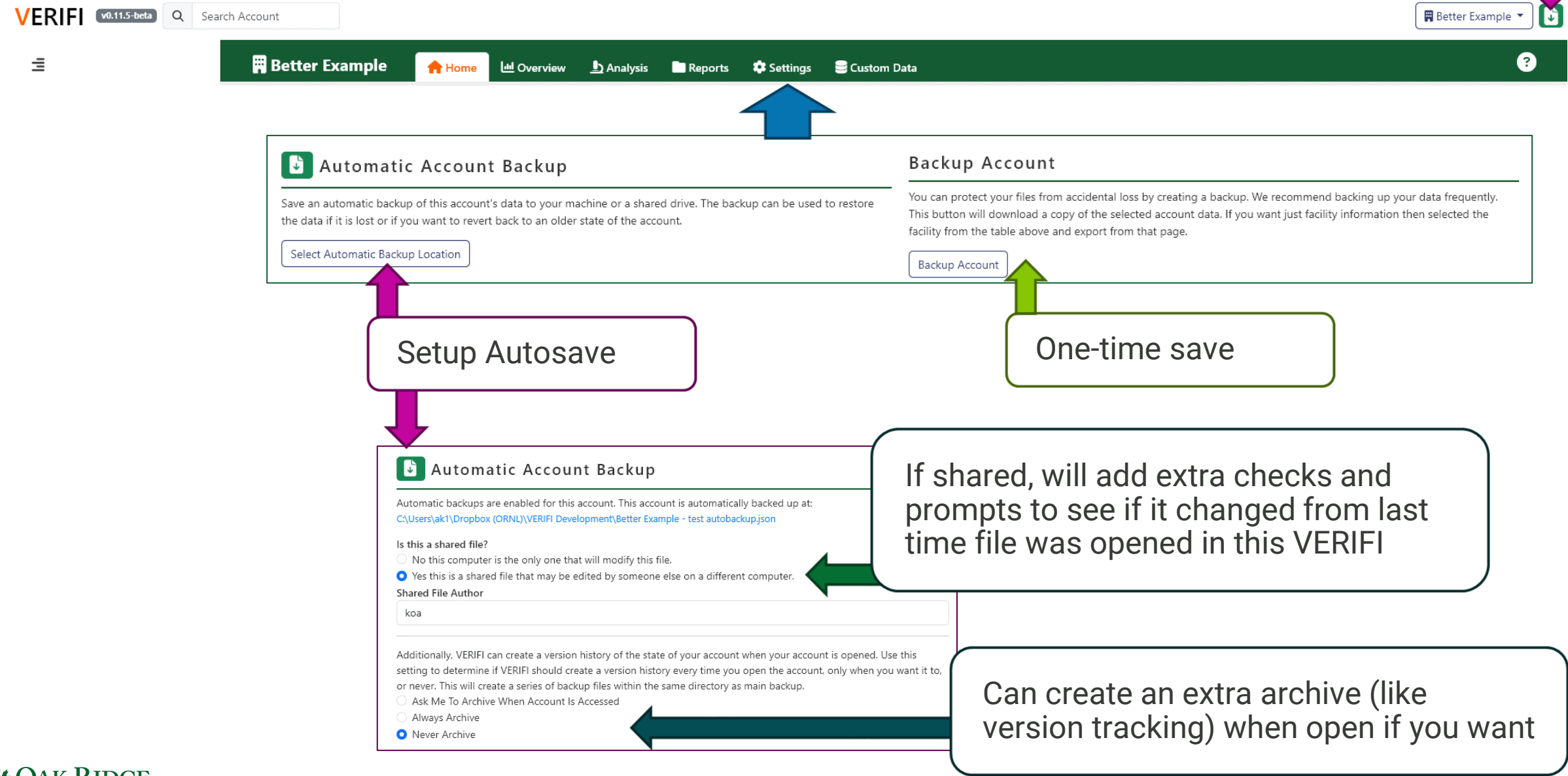


The screenshot displays the VERIFI web application interface. The main content area shows a table titled 'Monthly Analysis' for the 'West' facility. The table lists monthly data from November 2021 to December 2023. A large green arrow points to the 'Copy Table' button located below the table.

Month	Year	Value 1	Value 2	Value 3	Value 4	Value 5	Value 6	Value 7	Value 8
November	2021	11,480	14,125	-425.84	---	-425.84	2,645.8	18.73 %	21.01 %
December	2021	12,531	14,815	502.76	---	502.76	2,284	15.42 %	21.15 %
January	2022	13,283	15,876	395.03	---	395.03	2,592.4	16.33 %	20.62 %
February	2022	12,152	14,565	840.68	---	840.68	2,413	16.57 %	20.14 %
March	2022	12,438	15,421	-237.83	---	-237.83	2,983.1	19.34 %	20.09 %
April	2022	11,420	14,345	-273.25	---	-273.25	2,924.8	20.39 %	20.07 %
May	2022	11,457	14,289	774.23	---	774.23	2,831.9	19.82 %	20.01 %
June	2022	10,704	13,048	118.35	---	118.35	2,344.2	17.97 %	19.72 %
July	2022	10,739	13,623	-220.7	---	-220.7	2,883.4	21.17 %	19.6 %
August	2022	10,990	13,740	-223.72	---	-223.72	2,750.4	20.02 %	19.64 %
September	2022	10,742	14,679	620.59	---	620.59	3,937.4	26.82 %	19.75 %
October	2022	11,126	14,596	-425.83	---	-425.83	3,470.1	23.77 %	19.67 %
November	2022	11,324	15,234	682.75	---	682.75	3,909.5	25.66 %	20.27 %
December	2022	11,939	14,205	-107.06	---	-107.06	2,265.5	15.95 %	20.33 %
January	2023	12,385	14,684	-796.94	---	-796.94	2,298.8	15.66 %	20.31 %
February	2023	11,701	14,456	731.72	---	731.72	2,754.8	19.06 %	20.52 %
March	2023	12,242	15,270	-389.59	---	-389.59	3,027.6	19.83 %	20.56 %
April	2023	11,102	13,519	-1,098.5	---	-1,098.5	2,416.9	17.88 %	20.36 %
May	2023	11,111	14,617	1,102.7	---	1,102.7	3,505.8	23.98 %	20.72 %
June	2023	10,704	13,307	376.78	---	376.78	2,602.8	19.56 %	20.84 %
July	2023	10,953	13,462	-381.08	---	-381.08	2,508.7	18.64 %	20.64 %
August	2023	11,085	14,066	101.74	---	101.74	2,980.2	21.19 %	20.73 %
September	2023	10,813	14,086	27,648	---	27,648	3,272.6	23.23 %	20.42 %
October	2023	11,171	14,495	-527.29	---	-527.29	3,324.5	22.94 %	20.34 %
November	2023	11,308	15,176	625.2	---	625.2	3,868.9	25.49 %	20.33 %
December	2023	12,148	14,343	30,755	---	30,755	2,194.9	15.3 %	20.27 %

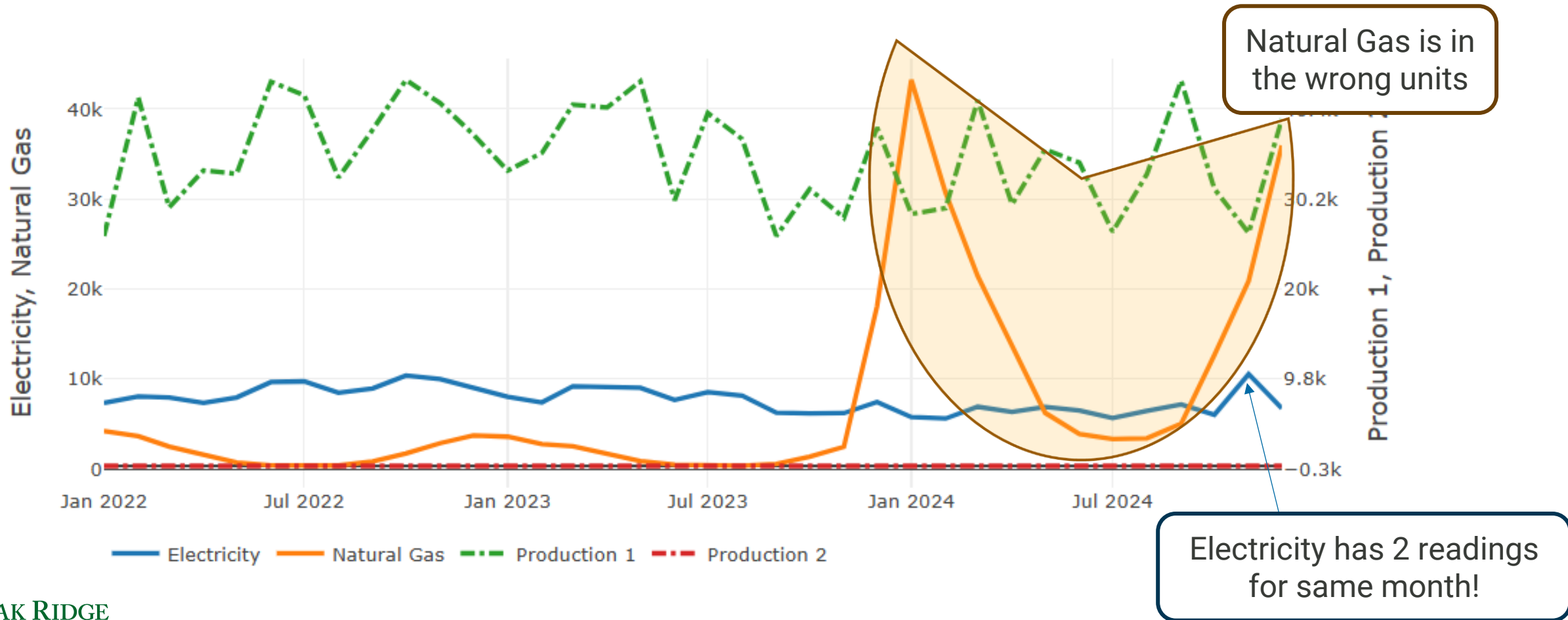
Autosave – Desktop Only

Autosave on & up to date



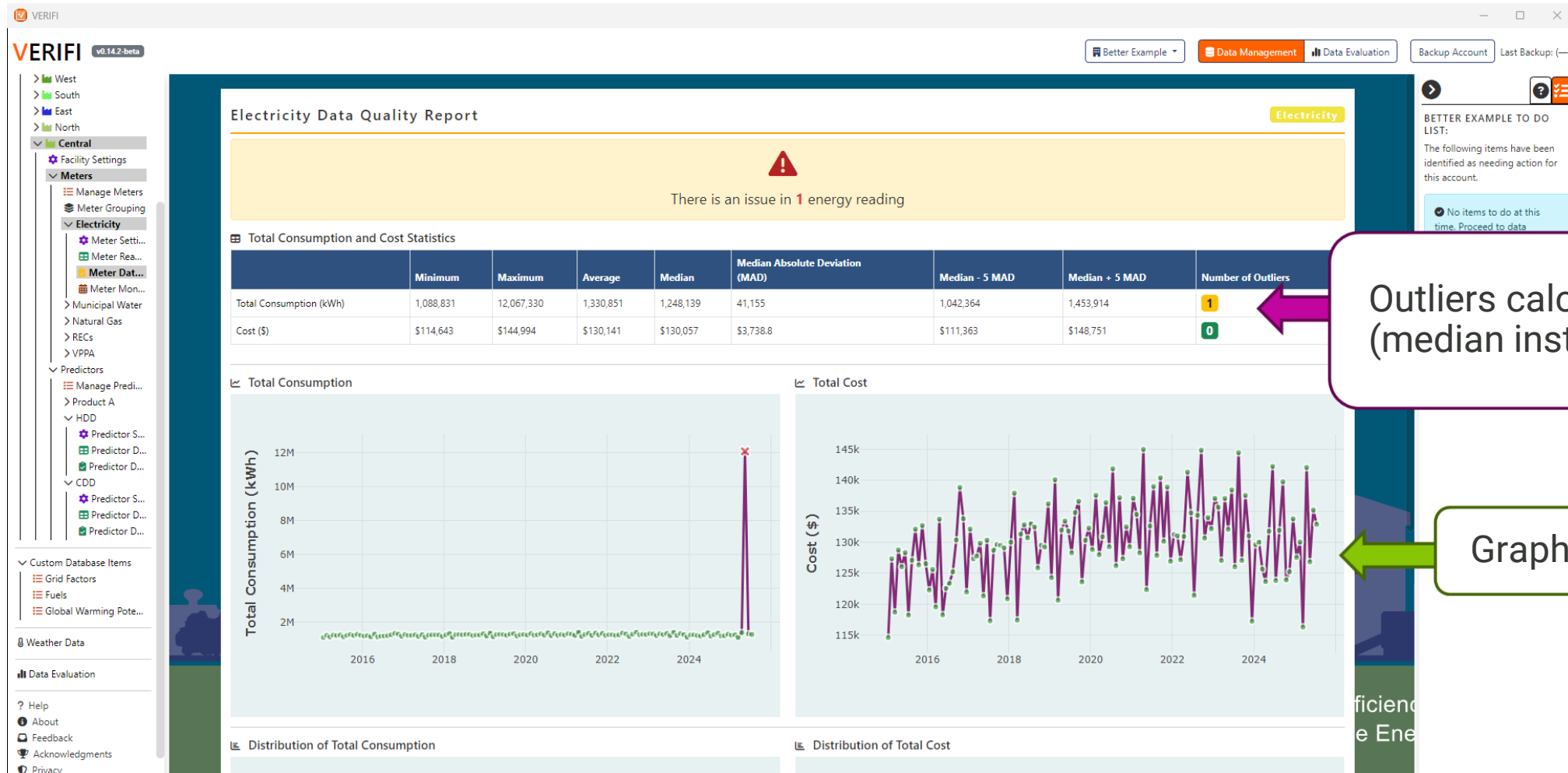
Something feel off with your data?

Check Data Visualization!



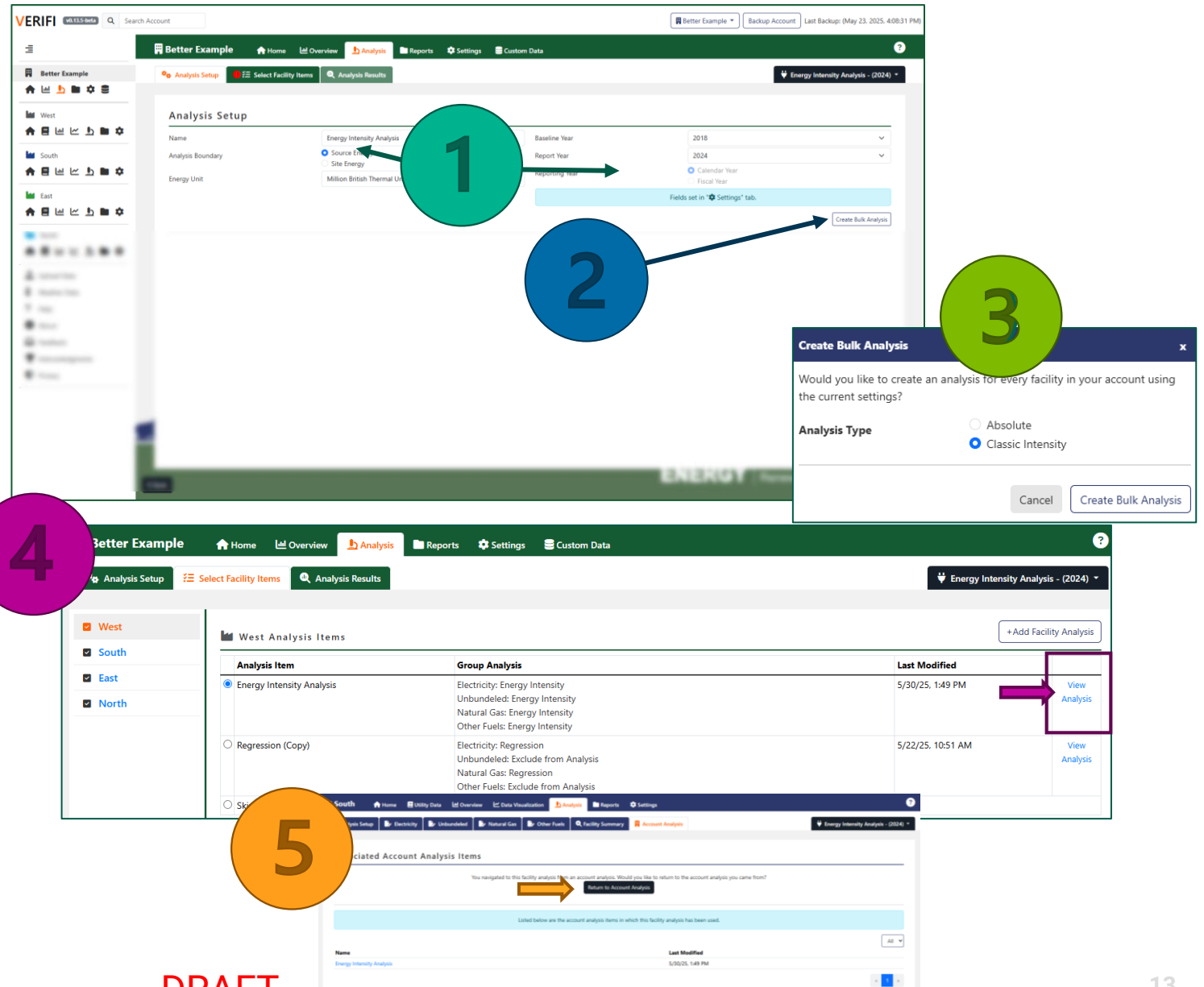
Something feel off with your data?

Check Data Quality Report



Just wanting Intensity or Absolute results? Bulk Analysis

1. Set up a **Company-level Analysis** (set year, give it a good name)
2. Click “Create Bulk Analysis”
3. Select Absolute or Classic Intensity
4. Review each facility, View  the facility analysis to modify
5. Return quickly under  “Account Analysis”



The screenshot displays the VERIFI web application interface for setting up and reviewing a Bulk Analysis. The interface is divided into several sections:

- Analysis Setup:** This section allows users to configure the analysis. It includes fields for Name, Analysis Boundary, Energy Unit, Baseline Year, Report Year, and Reporting Year. A green circle with the number 1 highlights the "Source Energy" dropdown menu. A blue circle with the number 2 highlights the "Create Bulk Analysis" button. A green circle with the number 3 highlights the "Create Bulk Analysis" button in the modal dialog.
- Create Bulk Analysis Modal:** This modal dialog asks if the user wants to create an analysis for every facility in their account using the current settings. It includes a section for "Analysis Type" with radio buttons for "Absolute" and "Classic Intensity". A green circle with the number 3 highlights this section.
- West Analysis Items:** This section displays a table of analysis items for the "West" facility. The table has columns for "Analysis Item", "Group Analysis", and "Last Modified". A purple circle with the number 4 highlights the "View Analysis" link for the "Energy Intensity Analysis" item.
- Account Analysis:** This section displays a table of account analysis items. An orange circle with the number 5 highlights the "Return to Account Analysis" button.

The interface also includes a sidebar with navigation links for Home, Overview, Analysis, Reports, Settings, and Custom Data. The top navigation bar includes a search account field and a user profile dropdown.

Understand Model Validity

Models must meet certain statistical criteria to be considered valid.

- $R^2 > 0.5$
- Model p-value < 0.2
- Variable p-values all < 0.2
- At least one variable p-values < 0.1

Invalid models are filtered out by default

Predictor means for the baseline and report year must fall within specified ranges

- ± 3 Standard Deviations of model mean, or
- Between the model minimum and maximum values

Models that fail this check are still displayed by default*

* If the report year fails and it is not a goal achieving year, this can be overlooked if it is likely the values will go back into range (like during COVID)

Select Modeling Method

Calculate Models

Max Variable Combinations

3

Update Models

Current Models Generated: 5/30/25, 2:24 PM

Show Statistically Invalid Models

Show Models that fail Data Validation

Regression Model

Analysis Variables

Predictor A

HDD Generated (65F)

CDD Generated (50F)

Use For Analysis

2024

Regression Coefficient

0.0014636317175381994

0

4.285040285182762

See statistically invalid models

Notes here are FYI.

Coeff < 0 , low adjusted R^2

Model	Year	p-Values	R2	R2	Value	Formula	Model Notes	Data Validation Failures
☒	2024	Predictor A: 0.08 CDD Generated (50F): 0.00	0.738	0.680	0.00	$7,360 + (0.00146 \cdot \text{Predictor A}) + (4.39 \cdot \text{CDD Generated (50F)})$	—	Predictor A failed in baseline year.
☐	2022	Predictor A: 0.01 HDD Generated (65F): 0.07 CDD Generated (50F): 0.02	0.753	0.661	0.01	$6,280 + (-0.00208 \cdot \text{Predictor A}) + (7.26 \cdot \text{HDD Generated (65F)}) + (12.9 \cdot \text{CDD Generated (50F)})$	Predictor A coef < 0	—
☐	2018	Predictor A: 0.02 CDD Generated (50F): 0.01	0.695	0.627	0.00	$3,860 + (0.00141 \cdot \text{Predictor A}) + (1.32 \cdot \text{CDD Generated (50F)})$	—	Predictor A failed in report year.

<Return to Model Selection

Model Year	Variable p-Values	R2	Adjusted R2	Model p-Value	Formula	Model Notes
2024	Predictor A: 0.08 CDD Generated (50F): 0.00	0.738	0.680	0.00	$7,360 + (0.00146 \cdot \text{Predictor A}) + (4.39 \cdot \text{CDD Generated (50F)})$	—

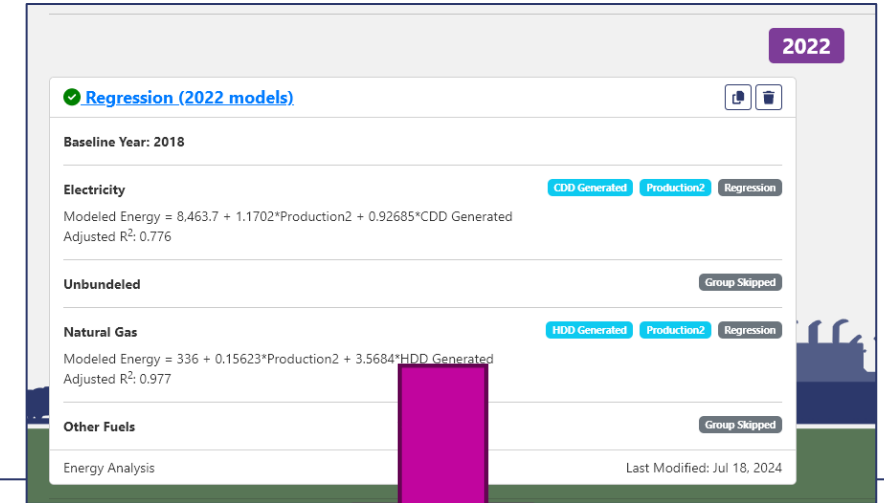
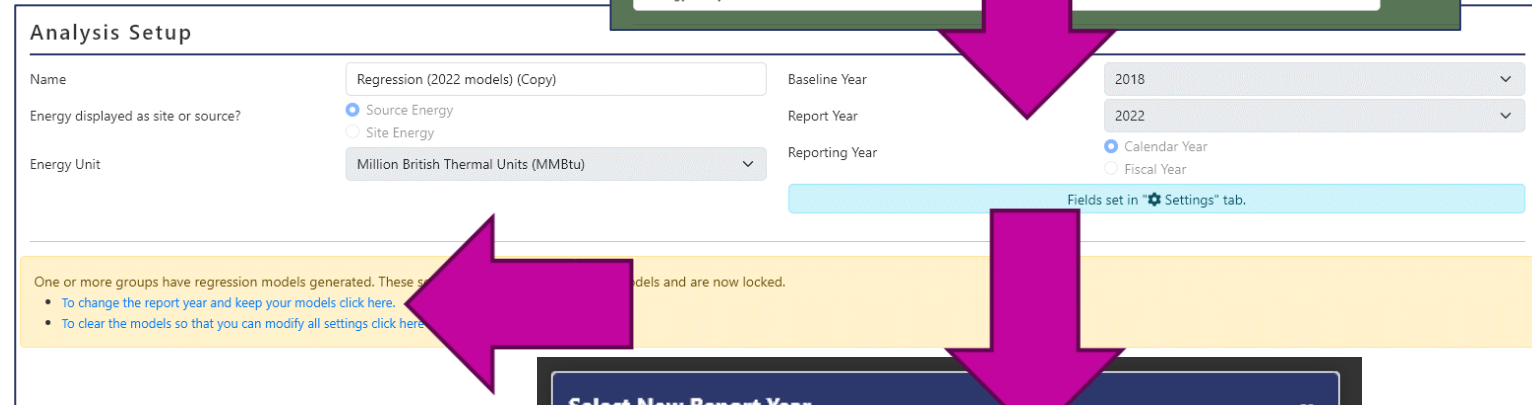
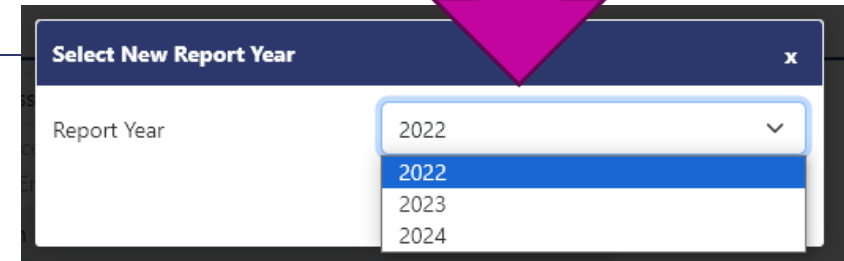
Predictor A CDD Generated (50F)

Mean Report Year Value	3,040,867	375.62
Mean Baseline Year Value	429,725	345.87
Range 1 Minimum of Model Variable	2,349,200	8.4167
Maximum of Model Variable	3,652,400	797.5
Range 2 Model Mean -3 Std Dev	1,601,340	-554.88
Model Mean +3 Std Dev	4,480,394	1,306.1
Validation Check	Fail	Pass

Learn more about model and see data validation here

Copy last year's analysis

- Go to Facility Analysis Page
- Click the  button in upper corner of the analysis card
- Select option in yellow box
- Select new year
- If using regression, check your models are still valid (data validation)

A screenshot of a web interface showing a "Regression (2022 models)" analysis card. The card has a purple header with "2022" and a green checkmark icon. It contains sections for "Electricity" and "Natural Gas", each with a regression equation and Adjusted R-squared value. There are buttons for "CDD Generated", "Production2", and "Regression" for Electricity, and "HDD Generated", "Production2", and "Regression" for Natural Gas. A "Group Skipped" button is also visible. The card is titled "Baseline Year: 2018".A screenshot of the "Analysis Setup" form. It includes fields for "Name" (set to "Regression (2022 models) (Copy)"), "Energy displayed as site or source?" (radio buttons for "Source Energy" and "Site Energy", with "Source Energy" selected), and "Energy Unit" (a dropdown menu set to "Million British Thermal Units (MMBtu)"). There are also dropdown menus for "Baseline Year" (set to 2018), "Report Year" (set to 2022), and "Reporting Year" (radio buttons for "Calendar Year" and "Fiscal Year", with "Calendar Year" selected). A light blue bar indicates "Fields set in 'Settings' tab." Below this is a yellow box with a message: "One or more groups have regression models generated. These models are now locked." and two links: "To change the report year and keep your models click here." and "To clear the models so that you can modify all settings click here." A large purple arrow points from the "Regression (2022 models)" card to the "Report Year" dropdown in the "Analysis Setup" form.A screenshot of a "Select New Report Year" dialog box. It has a dark blue header with the title and a close button. The "Report Year" field is a dropdown menu with a list of years: 2022, 2023, and 2024. The year 2022 is currently selected.

Performance Report

➡ Compare utilities, facilities, analysis groups!

➡ Understand the contribution each facility makes to your total savings!

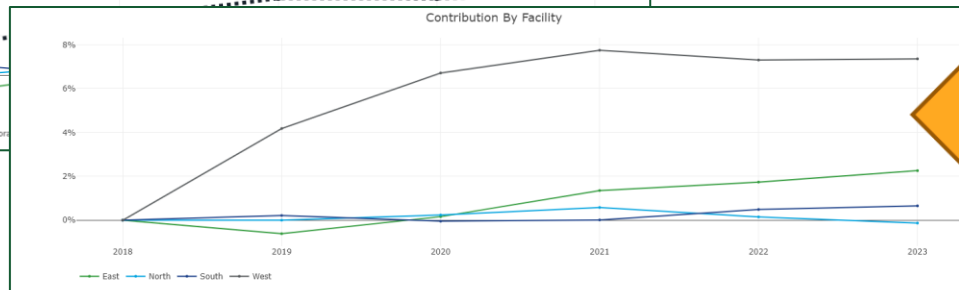
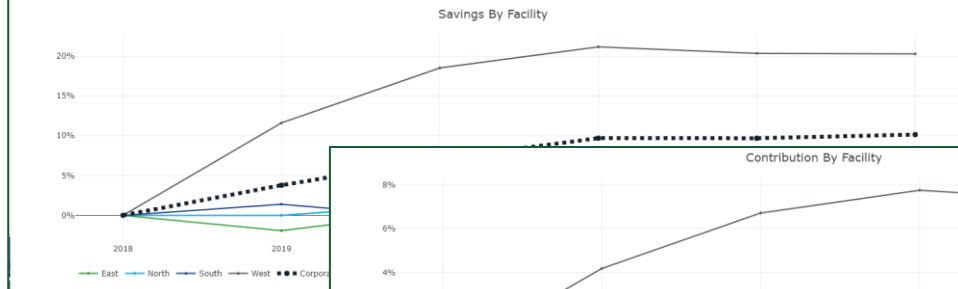
Facility Performance

Top 4 / Low 4 Performing Facilities

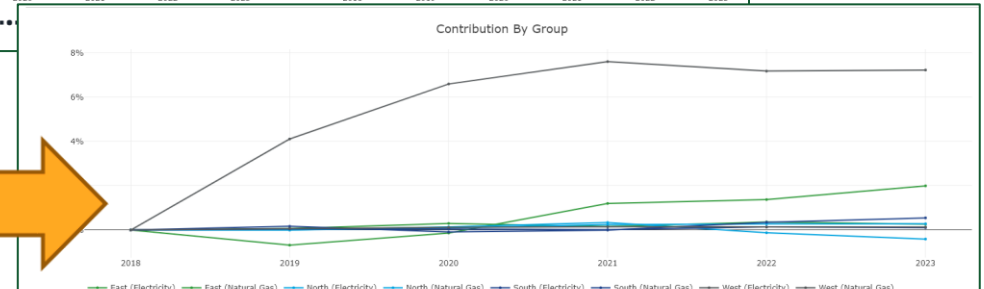
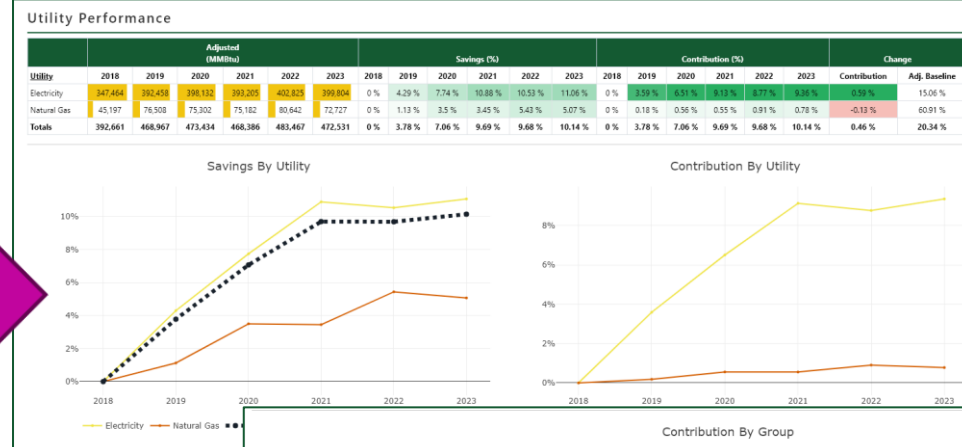
Change	Top 4	Savings	Low 4	Savings
1	West	20.27 %	North	-0.81 %
2	East	7.24 %	South	3.97 %
3	South	3.97 %	East	7.24 %
4	North	-0.81 %	West	20.27 %

Top 4 / Low 4 Change In Performance

Change	Top 4	Production	Low 4	Production
1	North	= %	West	-0.11 %
2	East	1.89 %	South	1.85 %
3	South	1.85 %	East	1.89 %
4	West	-0.11 %	North	= %



Facility	State	Adjusted (MMBtu)					Savings (%)					Contribution (%)					Change	
		2018	2019	2020	2021	2022	2018	2019	2020	2021	2022	2018	2019	2020	2021	2022	Contribution	Adj. Baseline
East	TENNESSEE	144,691	80,222	52,674	47,591	53,609	1.93	0.5	4.29	5.47	7.24	0	-0.61	0.16	1.35	1.74	0.51	1.89
North	MINNESOTA	0	78,988	77,916	74,770	79,596	0	1.44	3.61	0.91	-0.91	0	0	0.24	0.58	0.15	-0.13	= %
South	LOUISIANA	76,293	71,797	71,173	74,448	77,708	0	1.39	-0.3	0.07	3.08	3.97	0	0.21	-0.05	0.01	0.65	1.65
West	CALIFORNIA	173,677	145,960	116,671	111,577	111,630	11.6	18.51	21.15	20.31	26.27	0	-4.16	6.71	7.75	7.3	-0.11	
Totals		392,661	468,967	473,434	468,386	483,467	0	3.78	7.06	9.69	9.68	10.14	0	3.78	7.06	9.69	0.46	20.34



Facility	State	Utility	Adjusted (MMBtu)					Savings (%)					Contribution (%)					Change	
			2018	2019	2020	2021	2022	2018	2019	2020	2021	2022	2018	2019	2020	2021	2022	Contribution	Adj. Baseline
East	TENNESSEE	Electricity	10,365	26,783	19,264	23,877	28,628	-2.52	-0.5	4.53	5.16	7.5	0	-0.68	-0.14	1.2	1.37	1.99	0.62
East	TENNESSEE	Natural Gas	24,326	22,439	23,410	23,714	24,981	1.42	5.99	3.03	7.03	5.74	0	0.07	0.3	0.15	0.36	0.26	-0.11
North	MINNESOTA	Electricity	0	45,441	45,745	44,037	46,026	0	1.45	3.59	-1.32	-4.28	0	0	0.14	0.34	0.04	-0.13	-0.28
North	MINNESOTA	Natural Gas	0	33,547	32,171	30,733	33,989	0	1.43	3.68	3.95	4.26	0	0	0.1	0.24	0.27	0.28	= %
South	LOUISIANA	Electricity	69,811	65,180	65,474	68,076	69,402	1.2	-0.65	-0.03	2.43	3.59	0	0.17	-0.09	0	0.35	0.55	0.2
South	LOUISIANA	Natural Gas	6,482.4	6,616.8	5,699.1	6,371.9	7,239.7	3.25	3.72	1.07	9.33	8.91	0	0.05	0.04	0.01	0.14	0.11	-0.03
West	CALIFORNIA	Electricity	19,238	15,054	17,649	17,214	18,798	12.43	19.8	22.65	21.94	21.74	0	4.11	6.39	7.4	7.11	7.22	-0.19
West	CALIFORNIA	Natural Gas	14,389	13,906	14,022	14,363	14,852	2.36	-4	4.7	4.21	4.3	0	0.07	0.12	0.14	0.13	0.13	0.73
Totals			392,661	468,967	473,434	468,386	483,467	0	3.78	7.06	9.69	9.68	10.14	0	3.78	7.06	9.69	0.46	20.34

Adding new facilities to pledge?

- Keep your company baseline as it is
- Set the new facility's baseline to the first full reporting year with data
- Click the “This is a new facility” checkbox
 - This allows the facility analyses with a different baseline to appear in the company analysis
- That's it!

The image shows two overlapping screenshots of a web application interface for setting sustainability goals. The top screenshot is for a facility named 'Better Example' and the bottom is for 'North'. Both forms are titled 'Sustainability Questions' and contain several sections for setting goals.

Better Example Form:

- 1. Do you have Energy Reduction goal?** (checked)
 - Reduction Goal: 25 %
 - Baseline Year: 2018
 - Target Year: 2028
 - Is goal for absolute energy or energy intensity? (Intensity selected)
- 2. Do you have a Greenhouse Emission/Carbon Reduction goal?** (checked)
 - Reduction Goal: 50 %
 - Baseline Year: 2020
 - Target Year: 2028
 - Is goal for absolute GHG/Carbon or GHG/Carbon intensity? (Absolute selected)
- 3. Do you have Renewable Energy goal?** (unchecked)
- 4. Do you have a Waste Reduction goal?** (unchecked)
- 5. Do you have a Water Reduction goal?** (checked)
 - Reduction Goal: 25 %
 - Baseline Year: 2018

North Form:

- 1. Do you have Energy Reduction goal?** (checked)
 - Reduction Goal: 25 %
 - Baseline Year: 2019
 - Target Year: 2029
 - Is goal for absolute energy or energy intensity? (Intensity selected)
- 2. Do you have a Greenhouse Emission/Carbon Reduction goal?** (checked)
 - Reduction Goal: 50 %
 - Baseline Year: 2020
 - Target Year: 2030
 - Is goal for absolute GHG/Carbon or GHG/Carbon intensity? (Absolute selected)
- 3. Do you have Renewable Energy goal?** (unchecked)
- 4. Do you have a Waste Reduction goal?** (unchecked)
- 5. Do you have a Water Reduction goal?** (checked)
 - Reduction Goal: 25 %
 - Baseline Year: 2019
 - Target Year: 2029
 - Is goal for absolute water or water intensity? (Intensity selected)

At the bottom of the 'North' form, there is a checkbox labeled **This is a new facility** which is checked. A purple arrow points to this checkbox.